OpenEd Risks, Externalities, and Failure Modes

Purpose

This document lists what can fail, where, why, and how OpenEd should reduce the likelihood or impact.

Risk categories

1. Learning risk.
2. AI risk.
3. Assessment risk.
4. Trust/safety risk.
5. Security/privacy risk.
6. Market risk.
7. Operational risk.
8. Accessibility risk.

Risk register

Risk	Likelihood	Impact	Mitigation
Learners churn after first lesson	High	High	Improve onboarding, lesson variety, tutor, visual excitement.
Lessons feel like static content	High	High	Use lesson variants, practice, artifacts, visual explainers.
AI tutor hallucinates	Medium	High	Source grounding, citations, fallback, reporting.
AI tutor enables cheating	Medium	High	Assessment mode, hints first, artifact process prompts.
BYOK is confusing	Medium	Medium	Session default, simple copy, mock mode.
API keys leak from browser	Medium	High	Warn users; do not store by default; optional gateway; no logs.
Educators publish low-quality courses	High	High	Publish-readiness QA and review.

Risk	Likelihood	Impact	Mitigation
Review queue bottlenecks	Medium	Medium	Automated QA, risk triage, templates.
Rubric feedback feels unfair	Medium	Medium	Show criteria/examples, allow revision, educator review later.
Certificates become weak signals	Medium	High	Use proof ladder and portfolios.
Dark theme hurts usability	High	Medium	Light-first default, calmer dark mode.
Mobile experience is compromised	Medium	High	Design mobile flows first for learner app.
Source links break	Medium	Medium	Link checker and source metadata QA.
External videos blocked	High	Medium	Text/transcript fallback and reference media model.
Platform costs grow	Medium	High	BYOK, caching, managed AI limits later.
Legal/IP issues from course content	Medium	High	Source/license checks and educator warranties.
Harmful content appears	Medium	High	Safety policy, moderation, restricted topics review.
AI overreliance reduces learning	Medium	High	Socratic mode, retrieval practice, reflection.

Failure mode analysis by product area

Learner App

Failure points:

- Learner does not understand course value.
- Learner cannot set up BYOK.
- Learner does not know what to do next.
- Learner skips practice.
- Learner completes but has no proof.

Controls:

- Clear continue card.
- No-key tutor fallback.
- Short practice tasks.
- Artifact payoff.

- Portfolio preview.

Educator Studio

Failure points:

- Course builder too complex.
- Educator uses AI to produce generic content.
- Source mapping becomes burdensome.
- QA feels like punishment.

Controls:

- Guided/expert modes.
- Draft quickly, improve later.
- AI-assisted source cards.
- QA framed as publish readiness.

Assessment and Proof Engine

Failure points:

- Quizzes too easy.
- Artifacts too hard.
- Feedback too generic.
- Learners use AI to fabricate submissions.

Controls:

- Mixed question types.
- Examples and non-examples.
- Rubric specificity.
- Revision/process history.
- AI-use declaration.

AI Tutor

Failure points:

- Generic chatbot behavior.
- Incorrect answers.
- Direct answers during assessment.
- Prompt leakage.
- Data leakage.

Controls:

- Course context retrieval.
- Tutor modes.
- Assessment constraints.
- Server gateway for proprietary logic.
- Redaction and no-key-storage policy.

Externalities

Positive

- Free access to better learning.
- Lower anxiety due to private doubt-solving.
- Better educator tooling.
- More evidence-based skill development.

Negative

- Learners may outsource thinking.
- Educators may overproduce generic AI content.
- AI tutoring may widen gap between learners who can use keys and those who cannot.
- Artifact proof may create new pressure to perform.

Governance cadence

- Weekly: tutor failure review.
- Weekly: course QA queue review.
- Monthly: safety policy review.
- Monthly: metric review.
- Quarterly: market and competitor review.
- Quarterly: accessibility review.

Source notes

This document is grounded in the current OpenEd/FME project context and the following reference set where applicable:

- Uploaded Frontier Model Evaluations prototype code bundle, latest inspected bundle: Design System for Frontier Model Evaluation (5).zip.
- Uploaded Frontier Model Evaluations 51-hour curriculum and Module 1 foundations files.
- Coursera, Udemy, DataCamp, Moodle, Khanmigo, Duolingo public documentation and product pages.
- Supabase Auth and Row Level Security documentation.
- OpenAI API key safety guidance, MDN Web APIs, and YouTube IFrame Player API documentation.
- Bloom's 2 sigma problem, intelligent tutoring systems research, CourseAssist, LearnLM-Tutor, DCCI, and UNESCO guidance for generative AI in education.

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